



COPILOTTRAINING
TECHNICAL BRIEFING

GitHub Copilot CLI

| AI at the Point of Work

Close distance across intent, complexity, context, time, and geography.

GitHub Copilot CLI · Tech Talk · github.com/copilot



"How do I bring AI to where the work actually is — and keep steering it from wherever I am?"

Your day is 70% outside the editor. AI that stays in the IDE misses most of what's actually wrong.

The terminal sees everything. Now AI can too — and follow you home.

Software Developer

Debug, scaffold, and iterate without leaving the terminal or restarting context



DevOps Engineer

Triage CI failures, patrol infrastructure, and fix remote servers from any device



CLI Power User

Replace manual trial-and-error with Plan Mode and parallel agent execution



45 min → 8 min

Docker debugging with Plan Mode: 8 attempts down to 2 targeted checks

25 min → 5 min

CI build failure analysis automated via copilot - p in a GitHub Actions step

12 GB on the host

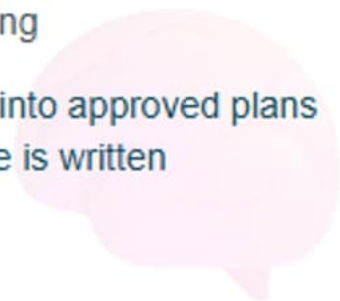
Log forensics performed where data lives — no scp, no compliance exposure

PART 1

Plan Mode

Clarify Intent Before Building

Turn ambiguous requests into approved plans before a single line of code is written



PART 2

Operating Modes

Choose Autonomy with Confidence

Interactive, programmatic, fan-out, delegated, and scheduled — with clear permission limits

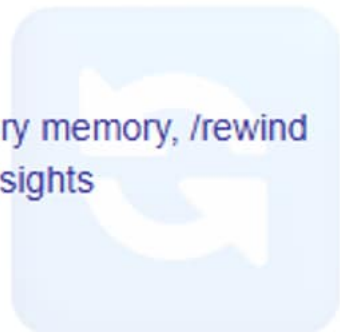


PART 3

Context Management

Keep the Session Alive

Auto-compaction, repository memory, /rewind recovery, and /chronicle insights

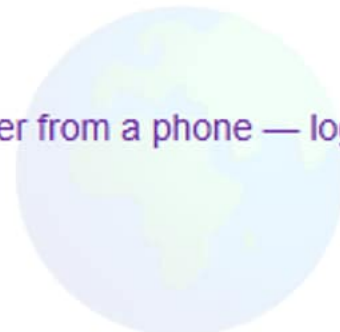


PART 4

Remote Sessions

The Last Distance Falls

Start on any machine, steer from a phone — logs stay where they live



Plan Before You Build

Open with the Distance Model, then close the intent gap through a quantified Docker story.



The Distance Model

Five gaps; Plan Mode closes intent first



Docker in 8 Minutes

Questions → plan → 2 checks vs 8 attempts



Rubber Duck Reviewer

Second model reviews plans — enabled by default

Traditional: 8 attempts, 45 minutes, wrong assumptions multiplied
↳ Plan Mode: 2 targeted checks, 8 minutes, root cause found

Five Distances Copilot CLI Removes

Each section of this talk closes a different gap between you and the work

Intent	AI stops guessing, starts asking — Plan Mode clarifies before acting	Plan Mode
Complexity	One task decomposes into parallel subtasks across independent agents	/fleet
Context	Session survives compaction and remembers decisions across sessions	Auto-compact
Time	Work outlives your attention — background delegation frees the terminal	& delegate
Geography	AI meets the problem where it lives — steer from phone or browser	--remote

The session is the unit of work, not the shell

Trial-and-Error vs. Collaborative Planning

Shift+Tab activates Plan Mode — AI asks before acting

✗ Traditional Workflow

- 1 **Make request**
Describe what you need
- 2 **AI assumes intent**
Picks one interpretation confidently
- 3 **Review output and fix**
Wrong assumption discovered too late
- 4 **Repeat until correct**
8 attempts on average

🔄 45 min, 8 failed attempts

✓ With Plan Mode

- 1 **Make request**
Describe what you need
- 2 **AI asks clarifying questions**
Surfaces ambiguities before acting
- 3 **Approve the plan**
Strategy confirmed before code is written
- 4 **Execute with confidence**
2 targeted checks — root cause found

✓ 8 min, done

8 debugging attempts → 2 with Plan Mode

Plan Mode in Action: Docker Debugging

Root cause in 8 minutes — not 45

```
$ copilot
> "Debug why the backend container won't start"
🤖 Copilot (Plan Mode):
  Before I start, a few questions:
  1. Check docker-compose config, logs, or both?
  2. Include environment variable analysis?
  3. Any recent changes to services or networking?
> "Start with logs, then config if needed"

🤖 Copilot: Running 2 targeted checks ...
✅ Port 5000 mapped to 5001 in docker-compose.yml - app expects 5000 internally
✅ Fix applied. Container starts successfully.
```

45 min, 8 attempts → 8 min, 2 checks

Choose the Right Mode

Build a decision ladder: interactive diagnosis, headless CI, parallel fan-out, and background delegation.



Interactive vs Headless

copilot for conversation; copilot -p for CI pipelines



/fleet Fan-Out

Decompose a plan into parallel subtasks in one approval



Scheduled Patrols

/every weekday — recurring agent without external cron

Manual CI triage: 25 minutes, context-switching, no pattern history
↳ copilot -p in Actions: 5-minute analysis artifact, zero human intervention

Match the Mode to the Task



Interactive

Unknown root cause, iterative diagnosis, collaborative debugging with context

`copilot`

Plan Mode available

Real-time steering



Programmatic

Headless CI/CD execution, structured output, zero human interaction required

`copilot -p`

Actions integration

`--allow-tool scoped`



Cloud Delegation

20+ min tasks — security audits, refactors. IDE and terminal stay completely free

`& prefix`

PR on completion

GitHub-hosted agent

Build Failure Analysis — Zero Human Intervention

.github/workflows/build.yml

- ```
- name: Analyze failure with Copilot CLI
 if: failure()
 env:
 GITHUB_TOKEN: ${{COPILOT_GITHUB_TOKEN}}
 run: |
 copilot -p 'Analyze the build failure.'
 --allow-tool shell(gh)
 --allow-tool shell(git)
 > failure-analysis.txt

- name: Post analysis as PR comment
 if: failure()
 run: gh pr comment --body-file failure-analysis.txt
```



### 5-Minute Analysis

25-minute manual triage becomes an automated 5-min artifact



### Scoped Permissions

--allow-tool limits the agent to gh and git only



### PR Comment Output

Structured analysis posted directly to the pull request

## /fleet Executes the Plan — Sandboxes Contain the Risk

### /fleet Fan-Out

#### Explicit decomposition

Orchestrator assigns subtasks to subagents in parallel

#### Context isolation

Each subagent has its own window — no pollution between tasks

#### Use after Plan Mode approves the strategy

#### Cost note

Each subagent makes independent LLM calls — check multiplier

`/fleet` implement all phases of this auth plan

### Containment Options

#### Local sandbox (`/sandbox enable`)

OS containment — no VM, included in Copilot seat

#### Cloud sandbox (`--cloud`)

Isolated GitHub-hosted environment — metered

#### `disableBypassPermissionsMode`

Enterprise policy — prevents `--yolo` in org environments

#### Never use `--yolo` on a remote production session

## The CLI as a Recurring Agent Runner

Schedule patrol prompts in plain language — no external cron required (experimental)

```
$ copilot --experimental
> "/every weekday at 9am"
 Summarize overnight PRs and post to Slack #dev-standup

> "/every 30min"
 Check pod health in staging.
 Alert via remote session if anything is unhealthy.

✓ Session-scoped - restarts with --resume; fires while session is running
✓ Pairs with --remote: patrol runs on the server, you steer from anywhere
```

*Requires /experimental on · minimum 10s · maximum 1 day*

# Context That Carries

Two ideas: the durable session and turn-level recovery — the foundation that makes --remote worth using.



## Infinite Sessions

Auto-compaction at 95% — sessions worth reconnecting to



## /rewind Recovery

Roll back a bad turn — Copilot edits undone, your edits kept



## /chronicle Insights

Sessions → standup, tips, better copilot-instructions.md

Context lost every 12-20 turns; restart and re-explain every session

↳ /resume from any machine — session, memory, and repo context travel with you

## The Session Is the Unit of Work — Not the Shell

### ∞ Auto-Compaction

#### **Fires at 95% token limit**

Compresses history in background — no interruptions

**Important decisions and facts persist; verbose output pruned**

#### **/resume from any machine**

Session + repo memory + context travel together

```
/context # token usage breakdown
/usage # session stats: duration, edits, ...
/compact # compress manually anytime
```

### ↺ /rewind Recovery

#### **Roll back to an earlier turn**

No Git required — works on any folder

#### **Conversation only**

Rewind the chat, leave files as they are

#### **Conversation + files**

Restore Copilot-changed files; skip files you edited

**Tracked across editing tools, shell commands, and sub-agents**

```
> /rewind
Pick turn before the bad refactor
Choose: Conversation + files
```

## /chronicle Turns Session History into Actionable Intelligence



### /chronicle standup

Generates a standup report from recent sessions — what you built, what you fixed



### /chronicle tips

Reviews your usage patterns and surfaces personalized improvement suggestions



### /chronicle improve

Analyzes repo sessions and suggests additions to `.github/copilot-instructions.md`



### Open-ended queries

Ask what you worked on, what you repeated, what you could automate

# Work from Anywhere

Geography was the last constraint. --remote removes it — AI goes to the machine, you steer from anywhere.



## Steer from a Phone

Start on the server, approve from GitHub Mobile mid-walk



## Data Stays Local

12 GB logs analyzed on the host — no scp, no compliance risk



## Composable

Plan Mode + /fleet + Memory all work inside --remote

```
SSH in, copy logs, paste into chat, lose context switching tools
↳ copilot --remote on the server - steer the live session from your phone
```

## Start on the Server — Steer from Anywhere

prod-server-3 (us-east-1)

```
SSH into the server
ssh ops@prod-server-3.us-east-1
```

```
Start a remote session
$ copilot --remote
```

```
🔑 Remote session started.
Monitor and steer from:
 https://github.com/copilot/sessions/abc123
```

```
Or enable mid-session:
> /remote
```

### See in Real-Time

Watch Copilot work from any browser or GitHub Mobile

### Approve from Anywhere

Grant or deny tool permissions from your phone mid-walk

### Steer and Resume

Inject prompts or /resume the session from a new machine

## Production Incident — Walking to a Meeting

The session lives on the server. You steer from your phone.

### ✗ Without --remote

- 1 **Alert fires at 2 PM**  
You're walking to a 2:05 meeting
- 2 **Wait until after meeting**  
30+ min of unresolved degradation
- 3 **SSH in from laptop**  
Re-establish context from scratch
- 4 **Manually scp 12 GB logs**  
Compliance and bandwidth risk



30+ min delay, compliance exposure

### ✓ With --remote

- 1 **Alert fires at 2 PM**  
You're walking to a 2:05 meeting
- 2 **Open GitHub Mobile**  
Live session URL — copilot is already on the server
- 3 **Approve tool calls from phone**  
Copilot runs /fleet log analysis
- 4 **12 GB stays on the host**  
Analysis runs where the data lives



Diagnosed in-meeting, data never moved

**--remote: execution locality + operator locality, no data transfer**

## --remote Composes with Every Capability



### Plan Mode + --remote

AI asks clarifying questions before touching a production server — safer phone approvals



### /fleet + --remote

Parallel subtasks across multiple environments from one approved plan on one host



### Scheduled + --remote

Infrastructure patrol on the server — fires on schedule, you review from any device



### Memory + --remote

/resume from a new device — repository memory and session context travel with you

## From Trial-and-Error Commands to AI Sessions That Follow the Work

### Before

Re-explain codebase conventions every session restart

SSH in, manually copy logs, paste into chat, lose context switching tools

One task at a time — terminal blocked during long-running agent work

Skip planning under deadline pressure and spiral into 8 failed attempts

### After

Repository memory carries conventions into every future session automatically

—remote puts AI on the server — steer log forensics from any device

& delegates 20+ min tasks to GitHub's agent — IDE and terminal stay completely free

Plan Mode turns 45 min of trial-and-error into an 8-min approved strategy

**8 min**

Docker debugging with Plan Mode (was 45 min, 8 attempts)

**5 min**

CI build failure analysis via copilot -p (was 25 min manual)

**0 scp**

12 GB log forensics performed on the host — data never moves

## ✓ WHAT YOU CAN DO TODAY

### Today



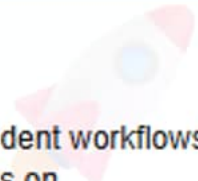
- ❑ Install Copilot CLI: `npm install -g @github/copilot`
- ❑ Press Shift+Tab before your next debugging session — activate Plan Mode
- ❑ Run `/rewind` after any wrong-turn refactor — no Git required

### This Week



- ❑ Add a `copilot -p` step to CI for automated build failure analysis
- ❑ Start a `--remote` session on a staging server and steer from GitHub Mobile
- ❑ Run `/chronicle improve` to improve your `.github/copilot-instructions.md`

### This Month



- ❑ Replace log-copy incident workflows with `--remote` sessions on production hosts
- ❑ Use `/fleet` after Plan Mode to parallelize your largest recurring refactors
- ❑ Enable `/experimental` and set up a `/every weekday` patrol for overnight PRs

### 🔑 Key Takeaway

The session is the unit of work — start on any machine, finish from wherever you are.

 **Core Documentation**[About GitHub Copilot CLI](#)

Core concepts, session model, and operating modes

[Install Copilot CLI](#)

Setup instructions for all platforms

[CLI Command Reference](#)

Full command syntax including `/rewind`, `/fleet`, `/chronicle`

[Steering a session remotely](#)

Remote session setup and mobile steering

 **Related Talks**[Copilot Memory](#)

Repository memory and cross-session context in depth

[Agent Plugins 1.0](#)

Portable skills, MCP config, and custom agents for the CLI

[CLI + Azure MCP](#) **How-To Guides**[Scheduling prompts](#)

`/every` and `/after` — recurring agent workflows

[Running tasks in parallel with `/fleet`](#)

Fan-out orchestration and subagent management

[Cloud and local sandboxes](#)

`/sandbox enable` and `--cloud` for containment

[Using `/chronicle`](#)

Standup, tips, and improve `copilot-instructions.md`



# GitHub Copilot CLI

AI AT THE POINT OF WORK — CLOSE DISTANCE ACROSS INTENT, COMPLEXITY, CONTEXT, TIME, AND GEOGRAPHY

## 8 min

Docker root cause — Plan Mode, 2 checks vs 8 attempts

## 5 min

CI analysis — copilot -p automates a 25-min triage step

## 12 GB

Log forensics on the host — data stays, --remote travels

What's your longest recurring debugging or triage loop — and which mode would you reach for first?

## Interactive Mode — Agent Environment

|                            |                                                |
|----------------------------|------------------------------------------------|
| <code>/init</code>         | Initialize Copilot instructions for this repo  |
| <code>/agent</code>        | Browse and select from available agents        |
| <code>/skills</code>       | Manage skills for enhanced capabilities        |
| <code>/mcp</code>          | Manage MCP server configuration                |
| <code>/plugin</code>       | Manage plugins and plugin marketplaces         |
| <code>/instructions</code> | View & toggle custom instruction files         |
| <code>/env</code>          | Show loaded instructions, MCPs, skills, agents |

### What this gives you

Everything that shapes the agent's behavior in this session — instructions, custom agents, skills, MCP servers, and plugins.

### Start here

`/init` once per repo, then `/env` any time you wonder "what does the agent actually know right now?"

## Interactive Mode — Models & Subagents

|                         |                                        |
|-------------------------|----------------------------------------|
| <code>/model</code>     | Select AI model to use                 |
| <code>/autopilot</code> | Toggle autopilot mode                  |
| <code>/fleet</code>     | Parallel subagent execution            |
| <code>/delegate</code>  | Send session to GitHub → create a PR   |
| <code>/tasks</code>     | View & manage tasks (subagents, shell) |
| <code>/sidekicks</code> | View running sidekick agents           |

### Three execution gears

- **Interactive** — you approve each step
- **Autopilot** — agent runs until done
- **Fleet** — multiple subagents in parallel

### Hand-off vs parallelize

`/delegate` ships the work to GitHub coding agent;  
`/fleet` stays local and fans out subagents.

## Interactive Mode — Code

|                              |                                          |
|------------------------------|------------------------------------------|
| <code>/plan</code>           | Implementation plan before coding        |
| <code>/diff</code>           | Review the changes made in current diff  |
| <code>/pr</code>             | Operate on pull requests for this branch |
| <code>/review</code>         | Run code review agent on changes         |
| <code>/ide</code>            | Connect to an IDE workspace              |
| <code>/lsp</code>            | Manage language server configuration     |
| <code>/terminal-setup</code> | Configure shift+enter for multiline      |

### Plan → Code → Review

`/plan` drafts the approach, the agent codes, then `/diff` and `/review` let you inspect & refine before opening a PR.

### IDE bridge

`/ide` pairs the CLI with an open editor — file edits open as native diffs and selections flow back to Copilot.

## Interactive Mode — Session

|                       |                                         |
|-----------------------|-----------------------------------------|
| <code>/resume</code>  | Switch sessions (id / task id / name)   |
| <code>/session</code> | View & manage sessions                  |
| <code>/rename</code>  | Rename or auto-name current session     |
| <code>/context</code> | Show context window token usage         |
| <code>/usage</code>   | Session usage metrics & stats           |
| <code>/compact</code> | Summarize history to free context       |
| <code>/memory</code>  | Toggle cross-session fact recall        |
| <code>/rewind</code>  | Undo last turn + revert file changes    |
| <code>/share</code>   | Export to markdown / HTML / gist        |
| <code>/copy</code>    | Copy last response to clipboard         |
| <code>/remote</code>  | Toggle control from GitHub web / mobile |

### Context hygiene

Watch `/context` . When it's > 70% full, run `/compact` with focus instructions instead of starting over.

### Sessions are persistent

All sessions live in `~/.copilot/session-store.db` .  
`/resume` picks any one up — even days later, by name or ID prefix.

## Interactive Mode — Permissions

|                                   |                                             |
|-----------------------------------|---------------------------------------------|
| <code>/allow-all</code>           | Enable all permissions (tools, paths, URLs) |
| <code>/add-dir</code>             | Whitelist a directory for file access       |
| <code>/list-dirs</code>           | Show all allowed directories                |
| <code>/cwd</code>                 | Change or show working directory            |
| <code>/reset-allowed-tools</code> | Clear the list of allowed tools             |

### Three permission surfaces

- **Tools** — what the agent can invoke
- **Paths** — what files it can touch
- **URLs** — what domains it can reach

### Denial always wins

Deny rules override `/allow-all`. Safe pattern: allow broadly, deny specifically (e.g. `shell(git push)`).

## Interactive Mode — Help & More

|                             |                                         |
|-----------------------------|-----------------------------------------|
| <code>/research</code>      | Deep research via GitHub + web          |
| <code>/chronicle</code>     | Session history insights                |
| <code>/changelog</code>     | CLI version history (+ AI summary)      |
| <code>/new</code>           | Start a new conversation                |
| <code>/clear</code>         | Abandon session and start fresh         |
| <code>/keep-alive</code>    | Prevent system sleep during work        |
| <code>/voice</code>         | Dictation via Foundry Local             |
| <code>/theme</code>         | View or set color mode                  |
| <code>/streamer-mode</code> | Hide model names / quota (screen share) |
| <code>/experimental</code>  | Enable / disable experimental features  |
| <code>/feedback</code>      | Provide feedback about the CLI          |
| <code>/help</code>          | Show help for interactive commands      |

### Long-running work

`/research` spins up a deep-dive task; pair with `/keep-alive` so your laptop doesn't sleep mid-run.

### Demo & screen-share kit

`/streamer-mode` hides quota and preview model names; `/theme` swaps light/dark to match your slides.

## Non-Interactive Mode — Scriptable AI

# One-shot prompt, exits after response

```
copilot -p "Fix the failing test in auth.test.ts" --allow-all
```

# Quiet output – only agent response, no stats (great for piping)

```
copilot -p "Summarize this PR" --silent | pbcopy
```

# Autopilot – continues autonomously until done

```
copilot -p "Migrate all API routes to use the new auth middleware" --allow-all --autopilot
```

# Share session output as a GitHub gist

```
copilot -p "Audit this repo for security issues" --allow-all --share-gist
```

# Structured JSON output for CI pipelines

```
copilot -p "Run tests and report failures" --allow-all --output-format json
```

## Permissions — Granular Control

# Allow specific tool categories

```
copilot --allow-tool=write --allow-tool=shell
```

# Allow all git except push

```
copilot --allow-tool='shell(git:*)' \
--deny-tool='shell(git push)'
```

# Restrict to specific directories

```
copilot --add-dir ~/workspace --add-dir /tmp
```

# Allow GitHub API access

```
copilot --allow-url=github.com
```

# Enable everything - tools, paths, URLs

```
copilot --yolo
```

# Same as:

```
copilot --allow-all-tools \
--allow-all-paths \
--allow-all-urls
```

**--allow-all** shorthand for all three

**--yolo** alias for **--allow-all**

# Required for non-interactive (-p) mode

```
copilot -p " ..." --allow-all-tools
```

## `copilot init` — Bootstrap Your Repo

*Analyzing codebase structure ...*

*Detecting build system: npm*

*Detecting test framework: jest*

*Detecting language: TypeScript*

✓ Created `.github/copilot-instructions.md`

### What it generates

Analyzes your repo with read-only tools and writes `.github/copilot-instructions.md` with:

- Build & test commands
- Coding conventions
- Project structure overview
- Technology stack details

### Why it matters

Every future session in this repo automatically loads these instructions — no need to re-explain your project to Copilot.

## `copilot login` — Authentication

# Interactive browser flow (default)

`copilot login`

# GitHub Enterprise Cloud (data residency)

`copilot login --host https://example.ghe.com`

# Headless / CI – use env var instead of login

`COPILOT_GITHUB_TOKEN=github_pat_... copilot`

# Token precedence order:

`COPILOT_GITHUB_TOKEN`

`GH_TOKEN`

`GITHUB_TOKEN`

### Supported token types

- Fine-grained PATs (v2) with *Copilot Requests* permission
- OAuth tokens from GitHub Copilot CLI app
- OAuth tokens from GitHub CLI (gh) app

✗ Classic PATs (ghp\_...) are not supported

### Token storage

Stored in system credential store. Falls back to plain text in `~/.copilot/` if no credential store is available.

## `copilot plugin` — Extend with Plugins

# Browse the default marketplace

```
copilot plugin marketplace browse copilot-plugin
```

# Install from GitHub repository

```
copilot plugin install github/my-plugin
```

# Install from local directory

```
copilot plugin install ./my-local-plugin
```

# List installed plugins

```
copilot plugin list
```

# Update / uninstall

```
copilot plugin update my-plugin
```

```
copilot plugin uninstall my-plugin
```

### What plugins can add

- Additional skills & agents
- Git hooks
- MCP servers
- LSP servers

### Built-in marketplaces

```
copilot-plugins github/copilot-plugins
```

```
awesome-copilot github/awesome-copilot
```

## copilot help config — Key Settings

### Behavior

**model** - AI model (gpt-5.4, claude-sonnet-4.6, ...)  
**autoUpdate** - auto-download CLI updates (default: true)  
**memory** - agentic cross-session fact recall (default: true)  
**keepAlive** - prevent system sleep (off / on / busy)  
**experimental** - enable experimental features  
**renderMarkdown** - render markdown in terminal  
**respectGitignore** - hide gitignored files from @picker  
**includeCoAuthoredBy** - add Co-authored-by to commits  
**compactPaste** - collapse large pastes (>10 lines)  
**streamerMode** - hide model names & quota (for screen sharing)

### Terminal

**mouse** - enable mouse support in alt screen  
**beep** - beep when attention required  
**banner** - always / once / never

### Security

**allowedUrls** - pre-approved URL/domain list (supports \*.domain)  
**deniedUrls** - blocked domains (takes precedence)  
**trustedFolders** - folders with persisted read/exec approval

### Available models

claude-sonnet-4.6   claude-sonnet-4.5   claude-haiku-4.5  
claude-opus-4.8   claude-opus-4.7   claude-opus-4.6  
claude-opus-4.6-fast   claude-opus-4.5  
gpt-5.5   gpt-5.4   gpt-5.2  
gpt-5.3-codex   gpt-5.2-codex  
gpt-5.4-mini   gpt-5-mini

### Teams

**companyAnnouncements** - messages shown in banner on startup

## copilot help environment — Key Env Vars

### Authentication

`COPILOT_GITHUB_TOKEN` - PAT / OAuth token (highest precedence)

`GH_TOKEN` - GitHub CLI token (second)

`GITHUB_TOKEN` - CI token (third)

`GH_HOST` - GitHub Enterprise hostname

`COPILOT_GH_HOST` - Copilot-only host (overrides `GH_HOST`)

### Behavior

`COPILOT_ALLOW_ALL=true` - skip all confirmations

`COPILOT_MODEL` - override AI model

`COPILOT_AUTO_UPDATE=false` - disable auto-update

`COPILOT_HOME` - override `~/.copilot` directory

`COPILOT_OFFLINE=true` - no network (needs BYOK)

`COPILOT_CUSTOM_INSTRUCTIONS_DIRS` - extra `AGENTS.md` dirs

### BYOK - Bring Your Own Key

`COPILOT_PROVIDER_BASE_URL` - custom API endpoint (activates BYOK)

`COPILOT_PROVIDER_TYPE` - openai / azure / anthropic

`COPILOT_PROVIDER_API_KEY` - API key

`COPILOT_PROVIDER_BEARER_TOKEN` - bearer token (takes precedence over API key)

`COPILOT_PROVIDER_WIRE_API` - completions / responses (use responses for GPT-5)

`COPILOT_PROVIDER_AZURE_API_VERSION` - Azure API version

### Proxy & Output

`HTTP_PROXY` / `HTTPS_PROXY` - outbound proxy URL

`NO_PROXY` - bypass list (comma-separated)

`NO_COLOR` - disable color output

`PLAIN_DIFF=true` - disable rich diff

## BYOK — Bring Your Own Key

```
Ollama (local, no auth required)
COPILOT_PROVIDER_BASE_URL=http://localhost:11434/v1 \
COPILOT_MODEL=deepseek-coder-v2:16b \
copilot

Custom OpenAI-compatible endpoint
COPILOT_PROVIDER_BASE_URL=https://my-api.example.com/v1
COPILOT_PROVIDER_API_KEY=sk- ... \
COPILOT_MODEL=gpt-4 \
copilot

Azure OpenAI with deployment name
COPILOT_PROVIDER_TYPE=azure \
COPILOT_PROVIDER_BASE_URL=https://my.openai.azure.com \
COPILOT_PROVIDER_MODEL_ID=gpt-4 \
COPILOT_PROVIDER_WIRE_MODEL=my-gpt4-deployment \
copilot
```

### Provider types

**openai** – default; covers Ollama, vLLM, Foundry Local  
**azure** – Azure OpenAI (host URL only)  
**anthropic** – Anthropic API

### When to use

- Air-gapped or offline environments
- Enterprise contracts with specific model providers
- Local development with Ollama / Foundry Local
- Cost control with self-hosted models

## `/fleet` — Parallel Subagent Execution

*Refactor auth module, add unit tests,  
update API docs, run the linter.*

`/fleet`

✓ Plan - 4 subtasks identified

[subagent-1] Refactor auth module

[subagent-2] Write unit tests

[subagent-3] Update API documentation

[subagent-4] Run linter across repo

*Running in parallel...*

✓ [subagent-4] Linter: 3 warnings fixed

✓ [subagent-1] Auth refactored

✓ [subagent-3] 12 endpoints updated

✓ [subagent-2] 24 tests added, passing

✓ All subtasks complete

### How it works

- Main agent acts as orchestrator
- Decomposes prompt into independent subtasks
- Runs in parallel, each with its own context
- Manages dependencies; merges results

### Best used for

- Multi-file refactors + test generation
- Audits + fixes across modules
- Large dependency updates in monorepos

### Power combos

Shift+Tab → plan → `Accept + autopilot + /fleet`

`"Use GPT-5.3-Codex to refactor..."`

`"Use @test-writer to create tests..."`

## Where the Magic Lives — `~/.copilot/`

How Copilot remembers everything — in-session and between sessions

|                                                |                        |
|------------------------------------------------|------------------------|
| <code>agents/</code>                           | custom agents          |
| <code>skills/</code>                           | skill packs            |
| <code>ide/</code>                              | IDE bridge             |
| <code>logs/</code> <code>session-state/</code> | in-session state       |
| <code>pkg/</code>                              | CLI & auto-update      |
| <code>restart/</code>                          | restart staging        |
| <code>config.json</code>                       | settings               |
| <code>permissions-*.json</code>                | allow/deny rules       |
| <code>session-store.db</code>                  | past sessions (SQLite) |
| <code>command-history-*.json</code>            | prompt history         |

### In-session

- `session-state/` — context, tasks, subagent state
- `ide/` — live editor bridge
- `logs/` — execution logs

### Between-session

- `session-store.db` — all conversations, resumable
- `command-history-*.json` — up-arrow recall
- `permissions-*.json` — persistent approvals

### Portable capabilities

- `agents/` & `skills/` follow you everywhere
- Override with `$COPILOT_HOME`